Named Grid Covers under Row–Column Margins: Three Exactness Levels and Largest-Piece Localization

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Abstract

We study finite named grid-piece placements whose aggregate row and column margins agree with a target, while overlaps and uncovered cells are allowed inside the relaxed fiber. Exact covers form a distinguished sublocus. This gives three different questions: whether a margin fiber supports an exact cover, whether every margin realization is exact, and whether every realization can reach an exact cover by replacing exactly two named pieces at a time. After fixing a pose of a largest selected piece, complement area q confines every participating residual placement to at most q active rows, q active columns, and q^2 cells. We derive fixed-parameter algorithms for all three questions in (q, h), where h bounds the orientation group, and the explicit fiber bound

$$h(q+1)(hq^2)^q = 2^{O(q \log q)}$$

for fixed h. Source-locked finite libraries show that the three predicates are genuinely distinct. A bounded 72-variable example further separates connectivity of one nonnegative fiber, generation of the integer relation lattice, and Markov connectivity over all right-hand sides.

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1 Introduction

Tomographic reconstruction asks what can be recovered from projections rather than from direct cell data. In grid tilings the feasible objects are usually required to be disjoint exact tilings, possibly with typed projections or a fixed tile shape. This literature already contains reconstruction algorithms, hardness results, exact domino models, and heterogeneous or polyatomic formulations [1, 5, 6, 7, 9, 10, 11, 8]. We ask a different, deliberately narrow question. Suppose that one named copy of each selected piece is placed inside a target and only the *aggregate* row and column margins are required to match. Overlaps and holes are allowed at this relaxed stage. What does that margin condition say about exact covers, and how hard is it to recognize the possible failure modes?

The distinction matters because a correct aggregate margin vector need not certify exact coverage cell by cell. We separate three predicates:

1. *support*: a nonempty margin fiber contains an exact cover;
2. *purity*: every point of the margin fiber is an exact cover;
3. *repair*: every fiber component, under replacement of exactly two named placements, contains an exact cover.

Purity implies repair, and repair implies support, but neither implication reverses. The strictness is witnessed by complete finite certificates for three source-locked named libraries. These witnesses organize the problem; the logical implications themselves are elementary, and general reconfiguration from infeasible states is prior art [12].

Our main uniform result uses the *largest-piece residual area*

$$q = |T| - |L|,$$

where L is a largest selected piece. Once a legal pose of L is fixed, the residual row and column margins have total mass q. Consequently every residual placement that can occur in the margin fiber is confined to at most q active rows, at most q active columns, and hence at most q^2 active cells. A translation bound then gives at most $h(q+1)$ poses for the largest piece and yields a $2^{O(q \log q)}$ bound on the complete margin fiber for fixed orientation bound h. Enumerating that bounded fiber decides support, purity, and exact-two repair.

1.1 Contributions and scope

The paper makes the following precise statements.

- We define the named one-copy aggregate-margin fiber and its exact-cover sublocus, together with support, purity, and exact-two repair.
- We prove the active-carrier lemma and fixed-parameter tractability of all three predicates in (q, h) for finite cell-list pieces.
- We give an exact participation-type compression for existential decision and compressed classification in a finite named library, under an explicit residual-area threshold. Literal listing remains output-sensitive.
- We record a simple overlap energy whose strict descent is sufficient for repair, without claiming necessity.
- We use certified C32, C46, and P49 libraries to separate the three exactness levels.
- In one frozen P21 configuration we distinguish connectivity of a selected nonnegative fiber, integral relation-lattice generation, and all-right-hand-side Markov connectivity.

The only uniform mathematical headline is the active-carrier/FPT theorem. We do not claim to introduce heterogeneous tiling tomography, pair replacement, infeasible-state repair, or algebraic tiling connectivity. Locked pair phenomena and binomial move connectivity for exact tilings have already been studied [13, 14]. The algebraic language in the P21 section is classical [17, 18, 19]. No priority or "first FPT" claim is made.

1.2 Organization

Section 2 fixes the finite input contract and the exact-two graph. Sections 3 and 4 establish the hierarchy and its finite witnesses. Sections 5 and 6 prove localization and the FPT theorem. Section 7 treats implicit libraries and the overlap potential. Section 8 gives the bounded algebraic case study. Section 9 states the stop boundary. The appendices record the evidence and reproduction contracts.

2 Named exact covers and tomographic fibers

2.1 Finite placement model

Let $T \subset \mathbb{Z}^2$ be a finite target cell set. A library consists of nonempty, named, finite cell-list pieces. Legal poses are obtained from a supplied finite orientation group H, with $|H| \leq h$, followed by integer translations; every legal pose must be contained in T. Explicitly, each $g \in H$ acts by a lattice automorphism $g : \mathbb{Z}^2 \to \mathbb{Z}^2$, the input supplies the action (or names one fixed action), and the pose catalogue consists of the distinct cell masks $gP + t \subseteq T$. Repeated masks produced by symmetries are identified. Names remain distinct even if two pieces have the same shape or area.

Fix a selected named set

$$E = \{P_0, \dots, P_r\}, \qquad \sum_{i=0}^{r} |P_i| = |T|.$$

The area equality is part of the decision domain. Area-incompatible selections are rejected during preprocessing. Write $\mathcal{P}_i$ for the finite catalogue of legal poses of P_i.

For a cell set $A \subseteq T$, let $X(A)$ be its joint row-and-column margin vector. A one-copy placement tuple is

$$x = (A_0, \dots, A_r) \in \prod_{i=0}^{r} \mathcal{P}_i.$$

Its cell coverage is

$$c_x(u) = |\{i : u \in A_i\}|, \qquad u \in T.$$

Definition 2.1 (Margin fiber and exact-cover sublocus)**.** For the target margin $X(T)$, define

$$\mathcal{F}_E = \left\{ x : \sum_i X(A_i) = X(T) \right\}$$

and

$$\mathcal{T}_E = \{x \in \mathcal{F}_E : c_x = \mathbf{1}_T\}.$$

Thus $\mathcal{T}_E \subseteq \mathcal{F}_E$. Points of $\mathcal{F}_E \setminus \mathcal{T}_E$ may have overlaps and uncovered cells, but every chosen pose still lies in the target.

The row and column margins are aggregate: the contribution of a piece name is not separately observed. This differs from typed projections that record one margin vector per tile type. It also differs from exact tiling models in which disjointness is imposed before the projection constraints.

2.2 The replacement graph

Definition 2.2 (Exact-two replacement graph)**.** The graph $\mathcal{G}_E^{(2)}$ has vertex set $\mathcal{F}_E$. Two distinct tuples are adjacent precisely when they differ in exactly two named coordinates.

An edge is one graph step, although its endpoints have Hamming distance two as named placement tuples. We also use the explicitly labelled variant $\mathcal{G}_E^{(\le 2)}$, which adds edges between tuples differing in one named coordinate. These graphs need not agree. Every unqualified repair statement in this paper refers to $\mathcal{G}_E^{(2)}$.

Definition 2.3 (Three exactness predicates)**.** For every area-compatible selected set E, define the following three Boolean predicates:

- *support-exact* if $\mathcal{F}_E \neq \varnothing$ exactly when $\mathcal{T}_E \neq \varnothing$;
- *fiber-pure* if $\mathcal{F}_E = \mathcal{T}_E$;
- *repair-exact* if every connected component of $\mathcal{G}_E^{(2)}$ meets $\mathcal{T}_E$.

If repair-exactness holds, the repair radius is

$$\rho(E) = \max_{x \in \mathcal{F}_E} \operatorname{dist}_{\mathcal{G}_E^{(2)}}(x, \mathcal{T}_E).$$

Equivalently, a finite library has any one of these properties on a declared selection domain $\mathcal{D}$ when the corresponding predicate holds for every $E \in \mathcal{D}$. Finite-library tables in this paper always state their domain explicitly. Empty fibers satisfy all three predicates vacuously and have radius zero. If a component is disjoint from $\mathcal{T}_E$, its vertices are called *trapped*; they are not assigned a finite repair distance.

2.3 Input-size contract

The target and piece masks are explicit finite cell lists. The orientation group is supplied explicitly by its lattice action, or by a fixed named lattice action of size at most h. Translations and margin vectors use binary-encoded integer coordinates. The input length $|I|$ includes these lists and coordinates. The theorem does not address arbitrary continuous bodies, oracle-described shapes, unlimited tile multiplicities, or anonymous tile types.

3 Three levels of tomographic exactness

Proposition 3.1 (Exactness hierarchy)**.** *For every area-compatible selected set,*

$$\textit{fiber purity} \implies \textit{repair-exactness} \implies \textit{support-exactness.}$$

Proof. If $\mathcal{F}_E = \mathcal{T}_E$, every graph vertex is already an exact cover, so every component meets $\mathcal{T}_E$. If repair-exactness holds and the fiber is nonempty, choose any component; it contains a point of $\mathcal{T}_E$. The reverse implication $\mathcal{T}_E \neq \varnothing \Rightarrow \mathcal{F}_E \neq \varnothing$ always holds because $\mathcal{T}_E \subseteq \mathcal{F}_E$. The empty case follows from the conventions in definition 2.3. □

The value of proposition 3.1 is organizational rather than logical difficulty. Generic discovery of a feasible solution from an infeasible state under restricted modifications belongs to the wider reconfiguration literature [12]. Here the predicates are tied to one specific aggregate tomographic relaxation and one specific named replacement graph.

Proposition 3.2 (Strictness in locked finite libraries)**.** *Neither implication in proposition 3.1 reverses within the certified finite named-library class:*

1. *C46 is repair-exact with radius one but is not fiber-pure.*
2. *The P49 row of zero-based indices* $(4, 9, 10, 11)$ *is support-exact, but eight non-tiling nodes are trapped singleton components.*
3. *In a separate failure, the P49 row* $(0, 1, 3, 14)$ *has a nonempty margin fiber and no exact cover.*

C32 is a positive pure anchor.

Proof. All assertions follow from the exhaustive finite proposition in proposition 4.1; the certificate and independent replay contract is recorded in Appendix A. □

These examples prove non-collapse of the predicates in the locked model. They do not classify shapes, characterize repairability, or imply that impure fibers normally repair.

4 Certified foundational libraries

4.1 Evidence layers and label conventions

Each finite statement below is separated into four layers.

1. A source lock records the asset descriptor, signature, direct locator, access date, and SHA-256 digest. The rasters are not redistributed.
2. A discrete normalization records the target cells, named piece masks, coordinate convention, and full D_4 pose convention.
3. An exact-cover census gives all tilings for every positive area-compatible row and exhaustive zero certificates for the rest.

4. An independent tomographic replay reconstructs every margin fiber, exact-cover subset, and exact-two graph downstream of the frozen normalized transcription.

The finite libraries C32, C46, and P49 are normalized from 4×4 geomagic specimens created and published by Lee Sallows [2, 3]. The piece geometry, targets, and displayed arrangements are Sallows' source material; Peter Cameron's finite group-action formulation is prior conceptual context [4]. The present project's contribution begins with the source-locked discrete normalization and consists of exhaustive pose and exact-cover censuses, aggregate-margin fibers, replacement graphs, and the uniform largest-piece-localization theorem. We do not claim authorship of the underlying geomagic specimens. The 86-row additive carrier shared by C32 and C46 is imported and is not presented as a new additive theorem.

C32 and C46 have distinct piece areas $1, \ldots, 16$, so a selected row lists both labels and areas. P49 has repeated areas; its row tuples always list zero-based locked piece indices. In particular, $(4, 9, 10, 11)$ has areas $\{4, 13, 1, 14\}$, whereas $(0, 1, 3, 14)$ has areas $\{15, 2, 3, 12\}$.

4.2 Source locks

C32 is the Sallows gallery asset *4x4 Durer5*, signed LS 11-06, with locked SHA-256

`c6093ccd58e725d6efde1750cccf47124a56d6a335fbe3c9b70c8ef66a9bf2cb`.

Its target is a 6×6 square with zero-based holes $(1, 1)$ and $(4, 4)$. Of twenty displayed panels, twelve already use these holes and eight use the horizontal reflection. Reflecting the latter group into the canonical target gives sixteen distinct labelled tilings.

C46 is the Sallows gallery asset *4x4 most perfect (2b)*, signed LS 4-09, with locked SHA-256

`86430abfcfe5793fecdedf42a84a0e4f08e16636e5691ab25cf4509e9c69ef00`.

It uses the same target and area-labelled piece convention. Its 36 displayed panels agree on the target mask and on the recovered connected color components.

P49 is the source-locked Sallows specimen *Art of Fugue*, signed LS 7-09, with locked SHA-256

`77efa6d693726aa026fce49e14480b933606ce552ff51a24eca453626d175c1e`.

Its target is a 6×6 square with the four corners removed. Its zero-based area sequence is

$$(15, 2, 12, 3, 4, 11, 7, 10, 4, 13, 1, 14, 9, 6, 12, 5).$$

Proposition 4.1 (Complete finite panel)**.** *Under the locked models and the semantics of Section 2, the complete area-compatible carriers have the following exact behavior.*

Library	*Rows*	*Tile/zero*	*Margin*	*Fiber = tile+extra*	*Support*	*Pure*	*Repair*
C32	*86*	20/66	*20*	$136 = 136 + 0$	*yes*	*yes*	*yes,* $\rho = 0$
C46	*86*	44/42	*44*	$352 = 344 + 8$	*yes*	*no*	*yes,* $\rho = 1$
P49	*94*	60/34	*61*	$1344 = 1208 + 136$	*no*	*no*	*no*

Here *Margin* is the number of nonempty margin fibers, while *Tile/zero* and the displayed fiber decomposition report the corresponding tiling counts and exhaustive zero certificates. Thus no positive exactness label is stated without accompanying existence data.

Proof. For each locked catalogue, the primary enumerator exhausts every area-compatible row and stores complete labelled exact covers or a zero certificate. A separately implemented replay regenerates D_4 poses, constructs the fibers by direct $2+2$ meet-in-the-middle matching, and agrees on every row count, edge count, component, radius, and verdict. Positive and negative controls pass, while the registered adversarial mutations are rejected. Exact commands and artifact paths appear in Appendices A and B. □

The replay is an independent downstream enumeration from these frozen normalized transcriptions. Because the original source rasters are not redistributed, the reviewer package does not independently reproduce image acquisition, segmentation, or transcription from Sallows' raster bytes. It verifies the normalized model and every mathematical deduction downstream of that source lock.

4.3 Witness rows

C32 has 136 margin nodes and every one is an exact cover. It is therefore pure. Seven of its positive rows are non-affine; this is useful as a warning against an affine-selection characterization but is not needed below.

C46 has 344 tilings and eight non-tiling margin matches. All eight extras occur in two rows:

Row	Fiber	Tilings	Extras	Components	Edges	Radius
$\{1, 3, 14, 16\}$	20	16	4	2	22	1
$\{2, 3, 13, 16\}$	8	4	4	4	4	1

Every extra is adjacent to a tiling. Hence C46 is repair-exact but not pure. The eight positive rows not displayed in the source are described only as *not source-displayed*: the source never claimed that its panel list was exhaustive.

P49 has two independent failure modes:

Indices	Fiber	Tilings	Extras	Components	Edges	Trapped comps./nodes
$(4, 9, 10, 11)$	24	8	16	16	8	8/8
$(0, 1, 3, 14)$	8	0	8	8	0	8/8

In the first row, eight extras pair with tilings and eight extras are *eight separate singleton components*. In the second row, all eight nodes are trapped singletons and no tiling exists. Across P49 there are sixteen trapped components. This corrects the inaccurate shorthand of one eight-node trapped component.

5 Largest-piece residual localization

Choose a largest selected name L, breaking area ties by a fixed name order, and put

$$q = |T| - |L|.$$

The residual pieces are nonempty and their total area is q, so there are at most q of them.

5.1 A translation cap

For finite nonempty $A, B \subset \mathbb{Z}^2$, define

$$D(A, B) = \{t \in \mathbb{Z}^2 : A + t \subseteq B\}.$$

Lemma 5.1 (Translation cap)**.** *For finite nonempty $A, B \subset \mathbb{Z}^2$,*

$$|D(A, B)| \leq \max(0, |B| - |A| + 1).$$

Equivalently, if $D(A, B)$ is nonempty, then

$$|D(A, B)| \leq |B| - |A| + 1.$$

Proof. The empty case is immediate. Suppose $D = D(A, B)$ is nonempty. Then $A+D \subseteq B$. Choose an integer linear functional ℓ injective on the finite set $A + D$; avoiding the finitely many forbidden integer slopes suffices. It is then injective on each factor as well. If $\ell(A) = \{a_1 < \cdots < a_m\}$ and $\ell(D) = \{d_1 < \cdots < d_n\}$, the chain

$$a_1 + d_1 < \cdots < a_m + d_1 < a_m + d_2 < \cdots < a_m + d_n$$

contains $m + n - 1$ distinct sums. Therefore

$$|B| \geq |A + D| = |\ell(A) + \ell(D)| \geq |A| + |D| - 1,$$

and rearrangement proves the claim. □

The sumset mechanism is classical; the lemma is included to make the algorithmic constants explicit. The maximum in the unconditional statement is essential when no translation exists.

5.2 The active carrier

Freeze one legal pose A of L and let

$$b = X(T) - X(A).$$

Since $A \subseteq T$, the vector b is nonnegative. Its row coordinates sum to q, and its column coordinates also sum to q.

Lemma 5.2 (Active carrier)**.** *After a pose A of L is fixed, every residual placement participating in a target-margin solution is confined to at most q active rows, at most q active columns, and therefore at most q^2 target cells.*

Proof. At most q row coordinates and at most q column coordinates of b are positive. If a residual pose Q occurs in a tuple whose residual margins sum to b, nonnegativity gives $X(Q) \leq b$ coordinatewise. Thus Q uses only rows and columns where b is positive. It is contained in

$$G_A = T \cap (\text{active rows} \times \text{active columns}), \qquad |G_A| \leq q^2.$$

□

The proof uses both containment of every pose in T and the declared row/column finite-grid margins. It is not a statement about continuous tomography.

Corollary 5.1 (Pose bounds)**.** *The largest piece has at most $h(q+1)$ legal poses. After one is fixed, each residual piece has at most hq^2 margin-compatible candidate poses.*

Proof. Apply lemma 5.1 to each oriented copy of L in T: every nonempty translation set has size at most $|T| - |L| + 1 = q + 1$. Sum over at most h orientations. Any residual pose compatible with the residual margin vector after A is fixed is a nonempty subset of the carrier G_A. The same lemma gives at most q^2 translations per orientation, and hence at most hq^2 candidate poses. This catalogue may overcount poses that do not extend to a full fiber tuple. □

6 Fixed-parameter decision and radius

6.1 Bounded dynamic programs

Fix a pose A of the canonical largest piece L, and retain the notation $b = X(T) - X(A)$. For margin feasibility, a partial-placement dynamic program needs only residual margin states bounded coordinatewise by b. Its exact state box has size

$$B(b) = \prod_j (b_j + 1).$$

For nonnegative integers n, $n + 1 \leq 2^n$. The row coordinates of b sum to q, as do the column coordinates, so

$$B(b) \leq 2^q 2^q = 4^q.$$

For exact coverage, the residual target $T \setminus A$ has q cells. A subset-mask dynamic program therefore has at most 2^q states. Disjoining over the poses of L decides whether the margin fiber and the exact-cover sublocus are empty.

To decide purity and repair, we reconstruct all accepting margin paths. This is safe because the entire fiber is itself parameter-bounded.

Theorem 6.1 (Largest-piece residual-area FPT theorem)**.** *For an area-compatible fixed selected set of named finite grid pieces and a finite orientation group of size at most h, support-exactness, fiber purity, and repair-exactness in the exact-two graph are fixed-parameter tractable in (q, h). The full margin fiber satisfies*

$$|\mathcal{F}_E| \leq h(q+1)(hq^2)^q = 2^{O(q \log(qh))}.$$

For fixed h, this simplifies to $2^{O(q \log q)}$. The replacement graph can be constructed in

$$2^{O(q \log(qh))} \operatorname{poly}(|I|)$$

time, and hence in $2^{O(q \log q)} \operatorname{poly}(|I|)$ time when h is fixed.

Proof. By corollary 5.1, L has at most $h(q+1)$ legal poses. For a fixed largest pose, there are at most q residual pieces and each has at most hq^2 margin-compatible candidate poses. Hence

$$|\mathcal{F}_E| \leq h(q+1)(hq^2)^q.$$

This includes nonaccepting combinations in its upper estimate and is therefore valid for the fiber. The case $q = 0$ is handled directly.

Enumerating the parameter-bounded fiber and testing the cell coverage of each tuple decides purity. Support is decided by comparing nonemptiness of $\mathcal{F}_E$ and $\mathcal{T}_E$.

Hash all fiber tuples. For each tuple, choose two named coordinates, enumerate alternatives from their candidate-pose catalogues, and retain the resulting tuple precisely when it occurs in the hash table. There are at most polynomially many choices outside the parameter-bounded pose factors, so this constructs $\mathcal{G}_E^{(2)}$ in $2^{O(q\log(qh))}\,\mathrm{poly}(|I|)$ time. Mark its connected components and test whether each intersects $\mathcal{T}_E$. This decides repair-exactness. □

Remark 6.1. Adding the one-name cases constructs $\mathcal{G}_E^{(\le 2)}$ in the same FPT class. This does not identify the two graphs or transfer finite component counts between them.

Corollary 6.1 (Existence bound for repair radius)**.** *Every repair-exact fiber satisfies*

$$\rho(E) \le |\mathcal{F}_E| - 1 \le h(q+1)(hq^2)^q - 1.$$

Proof. In a component meeting $\mathcal{T}_E$, choose a shortest path from a vertex to a tiling. A shortest path is simple and hence has at most the component size minus one edges. □

The bound in corollary 6.1 is an existence bound. It is not asserted to be polynomial, linear, sharp, or representative of observed runtime.

7 Implicit libraries and a sufficient repair potential

7.1 Exact participation types

Theorem 6.1 concerns a fixed selected named set. To avoid conflating that problem with a search through a large library, fix the following separate parameterized problem. The input is a finite target T, a finite named cell-list library $\mathcal{Q}$, the explicit orientation action, an integer Q, and one of the three predicates in definition 2.3. The question is whether there is an area-compatible selected set $E \subseteq \mathcal{Q}$ whose canonical largest name $L(E)$ satisfies

$$|T| - |L(E)| \le Q$$

and whose margin fiber has the requested predicate. The classification version returns one compressed record per behavior class; it does not list all named selected sets.

Fix a candidate largest name L, order its legal poses as $\mathcal{A}_L = (A_1, \dots, A_s)$, and put $q_L = |T| - |L|$. A residual name P is *eligible for* L when $|P| < |L|$, or when $|P| = |L|$ and the fixed tie order chooses L before P. For $A_j \in \mathcal{A}_L$, define

$$b_j = X(T) - X(A_j), \qquad \mathcal{M}_j(P) = \{B \in \mathcal{P}_P : X(B) \le b_j \text{ coordinatewise}\}.$$

Every mask in $\mathcal{M}_j(P)$ lies in the active carrier G_{A_j}. Use the common labelled carrier

$$S_L = \bigcup_{j=1}^{s} G_{A_j}, \qquad |S_L| \le h(Q+1)Q^2.$$

Definition 7.1 (Exact participation type)**.** The exact participation type of a residual name P, relative to L, is

$$\tau_L(P) = \big(|P|, \mathrm{eligibility}, (\mathcal{M}_1(P), \dots, \mathcal{M}_s(P))\big).$$

Each placement is stored as its actual labelled cell mask in S_L, not as an index local to one scenario. Thus the record retains the identity of a placement mask that occurs for more than one largest-piece pose. Let $m_L(\tau)$ be the number of available residual names of type τ.

Lemma 7.1 (Type-isomorphism lemma)**.** *Fix L. Suppose two residual named multisets $(P_1, \dots, P_r)$ and $(P'_1, \dots, P'_r)$ admit a bijection σ with $\tau_L(P_i) = \tau_L(P'_{\sigma(i)})$ for every i. Replacing each name by its matched name and retaining the same labelled placement mask gives an isomorphism of margin fibers. This isomorphism maps exact covers to exact covers and is a graph isomorphism for both $\mathcal{G}^{(2)}$ and $\mathcal{G}^{(\leq 2)}$.*

Proof. For every ordered pose A_j of L, equality of types gives exactly the same allowed cell masks in every matched residual coordinate. Hence the coordinatewise renaming is a bijection of placement tuples and preserves the sum of margin vectors. Because the actual masks are retained, it also preserves the coverage function and therefore the exact-cover sublocus. Two tuples differ in a named coordinate before the renaming exactly when their images differ in the matched coordinate. This remains true when the largest pose changes: cross-scenario mask identity records which unchanged residual coordinates are literally the same placement. Hamming distance one or two, and therefore both replacement graphs, are preserved. □

Proposition 7.1 (Compressed finite-library decision)**.** *The existential problem above and its compressed behavior classification are FPT in (Q, h). For a candidate L, a compressed record contains L, an exact participation-type count vector $k = (k_\tau)_\tau$, the number*

$$\prod_\tau \binom{m_L(\tau)}{k_\tau}$$

of named residual selections represented by that vector, and the resulting fiber predicates and counts.

Proof. For each candidate L with $0 \leq q_L \leq Q$, the ordered list of largest poses and the carrier S_L have size bounded by a function of (Q, h). Area, eligibility, and the exact mask family for every ordered largest pose therefore yield only $f(Q, h)$ possible types. The selected residual names are nonempty and have total area q_L, so at most Q of them are chosen. We may truncate the available multiplicity of every type at Q, enumerate the type-count vectors satisfying

$$\sum_\tau k_\tau \ \text{area}(\tau) = q_L$$

and the eligibility rule, and run theorem 6.1 on one representative of each vector. The type-isomorphism lemma proves that this representative has exactly the same fiber, exact-cover sublocus, and move graphs as every named selection counted by the displayed product. Constructing the signatures for all candidate largest names is polynomial in the explicit input size; all superpolynomial enumeration is confined to a function of (Q, h). □

Example 7.1 (Why literal listing is output-sensitive)**.** Take a $1 \times 2Q$ target, one named Q-bar, and N named monominoes. There are $\binom{N}{Q}$ distinct valid named selections. Consequently the literal list of all selections cannot, in general, be produced in total FPT time independent of output size. No Delay-FPT enumerator is claimed here; the distinction follows the standard parameterized-enumeration vocabulary [21].

7.2 Overlap energy

For $x \in \mathcal{F}_E$, define

$$\Phi(x) = \sum_{u \in T} \max(c_x(u) - 1, 0).$$

Proposition 7.2 (Energy identity and strict descent). *For every margin node x,*

$$\Phi(x) = |T| - \left|\bigcup_i A_i\right| = \frac{1}{2}\|c_x - \mathbf{1}_T\|_1, \qquad 0 \le \Phi(x) \le q.$$

Moreover, $\Phi(x) = 0$ exactly when $x \in \mathcal{T}_E$. If every non-tiling node has an exact-two neighbor of strictly smaller energy, then the fiber is repair-exact and $\rho(E) \le q$.

Proof. The chosen pose areas sum to $|T|$, so

$$\sum_u c_x(u) = |T|.$$

The overlap mass $\sum_u \max(c_x(u) - 1, 0)$ is therefore the total area minus the size of the covered union. Also $\sum_u (c_x(u) - 1) = 0$: positive deviation equals uncovered mass, proving the ℓ_1 identity. The largest pose covers $|T| - q$ distinct cells, so at most q cells can be missing from the union, whence $0 \le \Phi \le q$. Energy zero is equivalent to unit coverage everywhere. Finally, repeated strict descent in the nonnegative integer Φ reaches zero in at most $\Phi(x) \le q$ steps. □

Strict descent is sufficient, not necessary. Certified P21 and P26 rows contain repairable nonzero local minima that require a nondecreasing or uphill step. Thus local minima do not characterize trapped components.

8 P21: three non-equivalent connectivity questions

The uniform theorem concerns one selected one-copy margin fiber. The P21 example is included to show what that fiber cannot certify. It distinguishes:

1. connectivity of one fixed nonnegative fiber;
2. generation of the complete integer relation lattice;
3. connectivity of every nonnegative fiber, equivalently the Markov basis requirement.

The first fixes one right-hand side. The second allows signed intermediate combinations. The third varies the right-hand side and must preserve nonnegativity throughout [17]. Algebraic formulations of grid-tiling flip ideals also precede this example [15, 14]; the claims below are only for the frozen 72-variable configuration.

8.1 Frozen configuration

The source-locked specimen is *4x4 square target*, signed LS 11-02, with SHA-256

`2248f6e9d0d4cb793ef2ff8069a37ae62ed67629cee6263a0f7071ae4a274120`.

Its target is the 5×5 square. We freeze the zero-based piece-index row

$$E = (4, 5, 12, 13),$$

whose areas are $(7, 6, 6, 6)$. Only placements participating in the selected margin fiber are retained. Their block counts are $8, 24, 16, 24$, for 72 variables.

For each participating placement P in named block i, the configuration matrix A has column

$$e_i \mid X_{\mathrm{row}}(P) \mid X_{\mathrm{col}}(P).$$

Thus A is a 14×72 integer matrix of rank 12 and

$$K = \ker_{\mathbb{Z}}(A)$$

has rank 60.

For $s = 1, 2, 3, 4$, let $M_{\leq s}$ be the integer span of all squarefree one-copy relations whose two placement tuples differ in at most s named piece positions. This *piece support* is not toric degree, Graver support, or a claim of Graver primitiveness.

8.2 The fixed 32-node fiber

The chosen fiber contains 32 nodes: 8 exact tilings and 24 non-tiling margin matches. Its complete support filtration is:

Family	New rels.	Lattice rank	rank K/M	Components	Verdict
$M_{\leq 2}$	144	34	26	16	16 trapped nodes
$M_{\leq 3}$	932	51	9	4	all reach tilings
$M_{\leq 3} + \langle O_{714} \rangle$	one orbit	54	6	1	connected
$M_{\leq 4}$	7,444	60	0	1	complete graph

The relation counts are new at the indicated exact support. The complete $M_{\leq 3}$ and $M_{\leq 4}$ catalogues contain 1,076 and 8,520 relations, respectively. At support at most two there are sixteen two-node components: eight pair a tiling with an extra, while eight contain the sixteen trapped nodes. At support three there are four eight-node components, but every node reaches a tiling. At support four the graph is complete.

Proposition 8.1 (One connected fiber does not generate the lattice)**.** *The family $M_{\leq 3} + \langle O_{714} \rangle$ connects the fixed 32-node fiber, but has lattice rank $54 < 60 = \operatorname{rank} K$.*

Proof. The order-eight target action partitions the 7,444 support-four relations into 1,119 complete orbits. Exactly 36 single orbit modules connect the four components of the support-three graph. Direct Hermite/Smith reduction gives rank 54, with torsion-free quotient rank six, for every one of these choices; O_{714} is the canonical stored witness. □

8.3 The missing integral channels

The quotient $Q = K/M_{\leq 3}$ is torsion-free of rank nine. Among the 1,119 support-four orbits, 87 cyclic orbit modules have quotient rank three. They fall into three exact integral families of sizes

$$6, \quad 3, \quad 78.$$

Exactly three complete target-action orbits are necessary and sufficient to generate Q integrally. Every minimum triple chooses one orbit from each family, giving $6 \cdot 3 \cdot 78 = 1,404$ minimum triples. The stored canonical triple is $(283, 603, 714)$, with determinant -1 in the quotient basis.

The fixed 32-node fiber activates $0, 0, 36$ orbits in these three families. It is blind to the first two necessary integral channels. These numbers are configuration-specific finite facts, not a general equivariant-basis theorem; compare the general lifting context in [20].

8.4 Lattice generation is not Markov connectivity

All 60 Smith factors for $M_{\leq 4} \subseteq K$ are one, hence

$$M_{\leq 4} = K$$

integrally. This permits signed decompositions of every relation. It does not imply that a decomposition can be ordered while staying nonnegative.

Proposition 8.2 (All-right-hand-side no-go)**.** *Although $M_{\leq 4} = K$, the squarefree one-copy support-at-most-four family is not a Markov basis for the frozen 72-variable matrix.*

Proof. The complete quadratic census has 218 indispensable binomials: 144 cross-block quadrics occur in the one-copy family, while 74 within-block quadrics do not. A canonical missing quadratic lies in a two-monomial fiber with block count $(2, 0, 0, 0)$. No one-copy cross-block move applies in that fiber, so it is disconnected. Indispensability is certified using the standard binomial-fiber criterion [18, 19]. □

A separate two-monomial fiber with block count $(0, 0, 0, 7)$ forces an indispensable binomial of toric degree seven. Therefore every Markov basis has degree *at least* seven. The bounded minimum-generator census is:

Toric degree	Betti fibers	Minimum generators forced
2	218	218
3	996	996
4	5,705	5,727

At least 6,941 minimal generators are forced through degree four. No upper bound of seven, exact Markov degree, complete basis, Graver basis, universal Markov basis, or blockwise lifting theorem follows.

9 Limits and open directions

The theorem closes the decision problem under a precise finite contract; it does not close the wider geometry or algebra. The following directions are open and logically independent of the present manuscript.

Sharper radius. Corollary 6.1 bounds the radius by the number of fiber states minus one. The certified libraries have radii zero or one whenever repair succeeds, and strict energy descent would give $\rho \leq q$. No structural or sharp worst-case radius theorem is known.

Structural shape criteria. The active-carrier theorem is algorithmic. The finite C32/C46/P49 panel does not yield forbidden-pattern or shape-theoretic characterizations of support, purity, or repair.

Enumeration. Exact participation types give FPT existential decision and compressed classification. A Delay-FPT algorithm for literal named outputs is not an implemented result.

Continuous geometry. The C34 route would require a separate transcription and a continuous or polygonal theorem. It is costly, uncertain, and absent from the dependency graph.

Algebraic completion of P21. The exact Markov degree, a complete Markov basis, four block-internal bases, normality or compatible-projection hypotheses, and a structured lift remain open. They are deliberately parked: none is required for theorem 6.1 and propositions 8.1 and 8.2.

Priority boundary. The primary-source audit located no parameter-preserving collision with the precise active-carrier theorem, but it was not a systematic MathSciNet/zbMATH and citation-forest search. Any novelty, firstness, or priority claim would require reopening that audit. The current wording makes no such claim.

9.1 Conclusion

Aggregate target margins contain enough information to define a useful relaxation, but not enough to collapse exact coverage, repair, and mere existence into one notion. Largest-piece residual area restores algorithmic control: after a largest pose is fixed, all residual activity lives in a parameter-bounded carrier. The finite libraries make the logical distinctions concrete, while P21 shows that even complete knowledge of one fiber does not substitute for relation-lattice or all-right-hand-side Markov analysis.

A Evidence and responsibility map

A.1 Finite libraries

Statement	Canonical repository evidence
C32 source and normalization	`registry/c32_source_lock.json`; `reports/C32_SOURCE_AUDIT.md`.
C32 complete census	`reports/C32_EXACT_CENSUS.md`; Package B certificate, replay, and mutation outputs.
C46 source and normalization	`registry/c46_source_lock.json`; `reports/C46_SOURCE_AUDIT.md`.
C46 complete census	`packages/package_a_c46_reproduction/realization/REPORT.md` and its certificate, verifier, and mutation results.
P49 source specimen	`packages/package_c_comparative_hypergraphs/specimens/SPECIMENS.md`.
Three-library fibers	`docs/FOUNDATIONAL_SECTION.md`; `results/coarea_fpt_analysis.json`.

The source records establish displayed geometry and arrangements. The shipped artifacts establish discrete normalization, exhaustive realization counts, margin fibers, graph components, and radii. Raw copyrighted raster files are intentionally absent; source-image audits require local bytes matching the recorded SHA-256 digests.

A.2 P21

Statement	Canonical repository evidence
Source identity and geometry	`packages/package_c_comparative_hypergraphs/specimens/p21_locked.json` and `packages/package_c_comparative_hypergraphs/specimens/SPECIMENS.md`.
Mathematical case study	`docs/P21_CASE_STUDY.md`.
Quotient and support filtration	`packages/package_e_obstruction_channels/xray_channel/robust_p21_quotient.py`, `degree3_augmentation.py`, and `support4_ceiling.py`.
Minimum orbit and Markov boundary	`verify_minimum_orbit_generators.py` and `verify_markov_no_go.py`.
Independent integrated replay	`packages/package_e_obstruction_channels/xray_channel/verify_orbit_markov_closure.py`.

A.3 Claim boundary

This paper asserts only the theorem and certified finite statements listed in `docs/CLAIM_LEDGER.md`, at their stated scopes. It makes no novelty, firstness, or priority claim. In particular, the exact P21 Markov degree and a complete Markov basis remain open.

B Reproducibility contract

All commands below are run from the repository root with Python in UTF-8, no-bytecode mode and the exact environment in `requirements.txt`.

B.1 Manifest and core replay

```
python -X utf8 -B scripts/verify.py --profile manifest
python -X utf8 -B scripts/verify.py --profile core
```

The core profile checks the complete frozen manifest, manuscript/PDF receipt, C32 and C46 realization certificates, the P21/P26/P49 specimen locks, both residual-area producers, an integrated P21 smoke replay, archive-level checksums, import closure, and repository policy.

B.2 Full mutation and P21 replay

```
python -X utf8 -B scripts/verify.py --profile full
```

The full profile adds all three mutation suites. Its integrated P21 verifier reconstructs the 72 participating columns, 7,444 support-four relations, 1,119 target orbits, rank-nine quotient, minimum triples, and Markov obstructions without importing the principal support-four analysis file. It then repeats the full replay in a temporary tree containing only the manifest allowlist; this detects hidden dependencies on virtual environments, caches, build debris, or unshipped files.

B.3 Manuscript verification

The manuscript static verifier checks the frozen section order, labels, citations, load-bearing theorem fragments, no-firstness boundary, and presence of the built PDF. The PDF is then reopened, text-extracted for sanity, rendered to PNG, and visually inspected page by page. Build and verification commands are in `paper/BUILD.md`.

The code and result filenames beginning with `coarea_` are stable legacy artifact identifiers. In the mathematical prose the parameter is called the largest-piece residual area.

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